

Equality Chains (EQUAL: block)

Example 1 : Base case - length of the empty sequence

(a)

Prove $length(\langle \rangle) = 0$.

$$\begin{aligned} &length(\langle \rangle) \\ &= 0 \end{aligned} \quad \text{[by definition of length]}$$

Example 2 : Inductive step - length of a non - empty sequence

(a)

Prove $length(s \frown \langle x \rangle) = length(s) + 1$.

$$\begin{aligned} &length(s \frown \langle x \rangle) \\ &= length(s) + 1 \end{aligned} \quad \text{[by definition of length and concatenation]}$$

Example 3 : Multi - step chain with justifications

$$\begin{aligned} &length(s \frown t) \\ &= length(s) + length(t) && \text{[by length - concatenation lemma]} \\ &= length(t) + length(s) && \text{[commutativity of +]} \end{aligned}$$